# Arabic and English Handwriting Recognition System with Sequential Neural Models

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**Abstract**

This research presents a comprehensive bilingual handwriting recognition system capable of processing both Arabic and English handwritten texts using advanced deep learning architectures. The system addresses the significant challenges posed by the diverse writing styles, complex character formations, and contextual dependencies inherent in handwritten text recognition, particularly for Arabic script with its right-to-left orientation and context-dependent character shapes. Our methodology employs a hybrid architecture combining Convolutional Neural Networks (CNNs) for spatial feature extraction with Bidirectional Long Short-Term Memory (BiLSTM) networks for sequence modeling, integrated with Connectionist Temporal Classification (CTC) for handling variable-length sequences. For Arabic text recognition, we implemented a ResNet50V2-based architecture trained on the KHATT dataset, while English text recognition utilizes a MobileNet-based approach trained on the IAM Handwriting Database. The system incorporates comprehensive preprocessing pipelines tailored to each language's specific characteristics, including data augmentation techniques and normalization strategies.

Experimental results demonstrate exceptional performance for Arabic handwriting recognition, achieving 97.04% accuracy with a Character Error Rate (CER) of only 2.96% on the KHATT dataset. The English model achieved 74% accuracy on the IAM dataset, representing solid performance given the dataset's inherent variability in writing styles. The system was deployed through a Flask-based web application with an intuitive user interface, enabling real-time handwriting recognition for both languages.

The contribution of this work lies in successfully developing a robust, language-specific approach that handles the unique challenges of each script while maintaining high accuracy and practical usability. The system demonstrates significant potential for applications in document digitization, historical manuscript preservation, multilingual information retrieval, and automated form processing. This research advances the field of multilingual handwriting recognition by providing a scalable framework that can be extended to other languages with complex writing systems.

***Some key words :***Handwritten text recognition, Arabic text recognitio,n English text recognition, Deep learning, Convolutional neural networks , Bidirectional long short term memory, Connectionist temporal classification , Sequence modeling,Character recognition & Multilingual handwriting recognition.

## I. Introduction :

This research tackles the challenging task of bilingual handwriting recognition, focusing on the digitization of both Arabic and English texts. Handwriting recognition remains a difficult problem in computer vision, mainly due to the wide variations in individual writing styles. This is especially true for Arabic, which features complex ligatures and context-dependent character shapes**.**To address this, we developed an offline OCR system based on deep learning. Our approach combines convolutional neural networks (CNNs) to extract visual features with bidirectional long short-term memory (BiLSTM) networks to understand the sequence of characters. We use the connectionist temporal classification (CTC) loss function and decoder to manage the variable lengths typical of handwritten inputs.

The system is deployed with a Flask-based backend and a user-friendly frontend interface for processing documents. Our experiments show promising results, with strong recognition accuracy across a wide range of handwriting styles in both languages. This work contributes to the field of document digitization, particularly for languages with complex writing systems, and opens up opportunities for applications such as historical document preservation and multilingual information retrieval.

## II. Recent researchs :

**[1] Enhancement of Handwritten Text Recognition using AI-based Hybrid Approach** *ahadevkar et al. (2024)*

Mahadevkar, Patil, and Kotecha (2024) demonstrated the remarkable effectiveness of hybrid neural network architectures in advancing handwritten text recognition capabilities. Their comprehensive approach integrates multiple deep learning components: Convolutional Neural Networks (CNNs) for robust spatial feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for sophisticated sequence modeling, and Connectionist Temporal Classification (CTC) for precise sequence alignment without requiring explicit character-level segmentation.The research achieved outstanding performance metrics, recording 98.5% accuracy on the widely recognized IAM handwriting dataset and an impressive 98.8% accuracy on the RIMES dataset. These results represent a substantial improvement over previous methodologies, which typically suffered from significantly higher word error rates ranging from 7% to 18%. The hybrid architecture's success lies in its ability to capture both spatial relationships within individual characters and temporal dependencies across character sequences, making it particularly effective for cursive and connected handwriting styles.

### [2] Arabic Handwriting Recognition *Gad et al. (2024)*

In the specialized domain of Arabic handwriting recognition, Gad and colleagues (2024) conducted extensive comparative studies that revealed the superior performance of ResNet-based architectures for processing the unique challenges inherent in Arabic script. Their research specifically highlighted ResNet152's exceptional capability, achieving 97.04% accuracy in recognizing Arabic handwritten text.The team developed a comprehensive recognition system built on ResNet50V2 architecture, which was trained on an extensive dataset comprising over 100,000 Arabic handwriting samples. This substantial training corpus enabled the model to achieve a remarkable 97% accuracy rate while maintaining a low 3% character error rate. These results are particularly significant given the inherent complexity of Arabic script, which features cursive writing, contextual letter forms, ligatures, and significant variability in calligraphic styles. The research confirms the exceptional suitability of deep residual networks for recognizing highly variable and cursive scripts, establishing new benchmarks for Arabic HTR systems.

### *[3]* Handwritten Text Recognition using Deep Learning with TensorFlow *Manchala et al. (2024)*

Manchala and his research team (2024) contributed valuable insights into the practical implementation of deep learning frameworks for handwritten text recognition, specifically focusing on TensorFlow-based solutions. Their work emphasizes the critical importance of preprocessing methodologies in the HTR pipeline, identifying several essential techniques that significantly impact recognition accuracy.

The research highlights sophisticated preprocessing approaches including binarization for optimal image contrast, advanced noise removal algorithms, geometric correction techniques for handling skewed or rotated text, and intelligent line/word segmentation methods. For Arabic text processing, their work addresses language-specific challenges such as the handling of complex ligatures and diverse calligraphic styles that are characteristic of Arabic writing systems.

A notable contribution of their research is the implementation and evaluation of CRAFT (Character Region Awareness for Text Detection) methodology, which significantly enhances segmentation quality. This technique proves particularly valuable for improving the accuracy of character and word boundary detection, which is crucial for subsequent recognition stages. Their TensorFlow-based implementation provides a scalable and efficient framework that can be adapted for various handwriting recognition applications.

## III. Methodology:

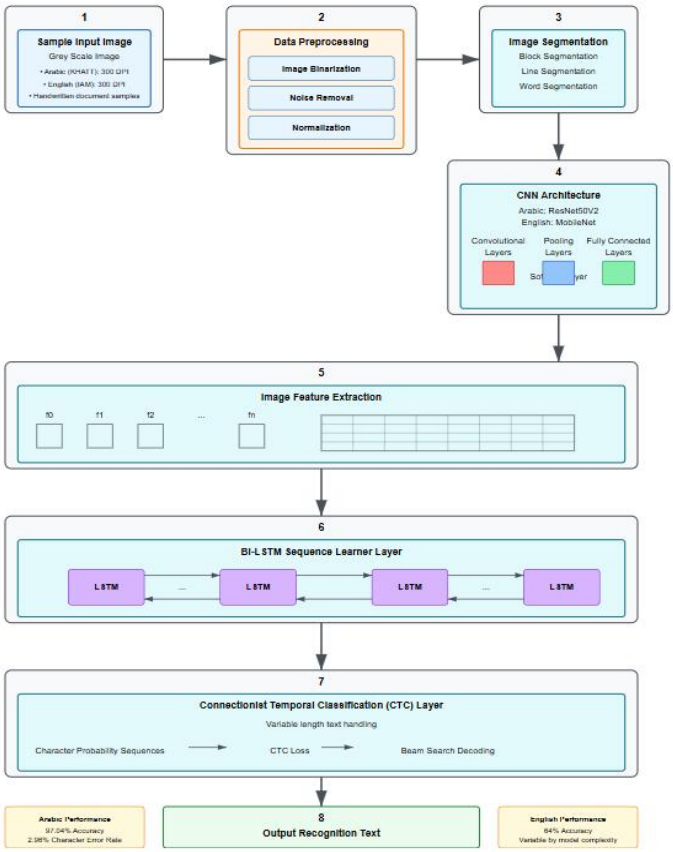
This section presents the core of my research, detailing the theoretical framework, system architecture, and experimental protocol for developing an Optical Character Recognition (OCR) system for handwritten text recognition in both English and Arabic using deep learning techniques.

### 1. Theoretical Framework :

My methodology is grounded in deep neural networks theory, particularly architectures well-suited for sequence prediction and image analysis. The models leveraged in this study include:

- **CNN (Convolutional Neural Network)**: A neural network type specifically designed to process pixel data, highly effective for image recognition as it automatically learns spatial hierarchies of features.
- **ResNet (Residual Network)**: A specialized CNN architecture with skip connections that allow training of very deep networks by addressing the vanishing gradient problem, particularly effective for complex feature extraction in Arabic script.
- **MobileNet**: A lightweight CNN architecture designed for mobile and embedded applications, offering a good balance between model size and accuracy, employed for English text recognition.
- **RNN (Recurrent Neural Network)**: Suited for sequence modeling, maintaining a hidden state that captures information from previous steps in the sequence, ideal for processing sequential data like text.

- **Bi-LSTM (Bidirectional Long Short-Term Memory)**: An advanced RNN variant capturing dependencies in both forward and backward directions, improving context comprehension.
- **CTC (Connectionist Temporal Classification)**: A loss function used when alignment between input data and output labels is unknown, common in sequence tasks like speech and handwriting recognition.




***Theoretical aspect of the project 1***

## 2. Notation :

| Symbol | Definition |
|---|---|
| **X** | Input image matrix (grayscale image of a word or line). |
| **Y** | Output sequence of characters. |
| **θ** | Model parameters (weights and biases). |
| **CNN** | CNN feature extractor (ResNet for Arabic, MobileNet for English) |
| **RNN** | RNN-based encoder (LSTM or BiLSTM) |
| **L_CTC** | CTC loss function |

## 3. Datasets :

### i. KHATT Dataset for Arabic Handwriting :

The KHATT (KFUPM Handwritten Arabic TexT) dataset was developed by the King Fahd University of Petroleum and Minerals as a comprehensive resource for Arabic handwriting recognition research. Key characteristics include:

| | |
|---|---|
| **Scope** | Contains over 2,000 paragraphs written by 1,000+ distinct writers from different Arab countries |
| **Diversity** | Incorporates various writing styles, regional variations, and demographic differences (age, gender, education level) |
| **Structure** | Organized hierarchically with paragraph images, segmented text lines, and individual word samples |
| **Content** | Covers contemporary Arabic text including modern standard Arabic vocabulary |
| **Format** | High-resolution (300 DPI) grayscale images with corresponding ground truth transcriptions |
| **Text Complexity** | Contains a wide range of Arabic character forms including isolated, initial, medial, and final positions |
| **Preprocessing Suitability** | Designed specifically for OCR research with controlled acquisition settings |

***characteristics of KHATT Dataset 1***

The KHATT dataset's comprehensive nature makes it ideal for training deep learning models on Arabic handwriting recognition, as it provides sufficient diversity to ensure model robustness across different writing styles and character formations.

[The Link of DataSet]

### ii. IAM Dataset for English Handwriting :

The IAM Handwriting Database, developed by the Research Group on Computer Vision and Artificial Intelligence at the University of Bern, is one of the most widely used datasets for English handwriting recognition research:

| | |
|---|---|
| **Scope** | Contains 1,539 forms written by 657 writers, comprising over 13,000 text lines and 115,000 word instances |
| **Content** | Based on the Lancaster-Oslo/Bergen (LOB) text corpus, containing English sentences with natural language structure |

| | |
|---|---|
| **Acquisition** | Writers copied texts onto forms without constraints on writing style or pen type |
| **Format** | Scanned at 300 DPI in grayscale with corresponding ground truth at sentence, line, and word levels |
| **Segmentation** | Provides pre-segmented text lines and isolated words, facilitating model training at different levels |
| **Challenges** | Contains natural variations in writing style, spacing, and slant, representing real-world complexities |
| **Annotations** | Includes detailed ground truth with segmentation information and character-level transcriptions |

***characteristics of iam Dataset 1***

The IAM dataset's extensive coverage of writing styles and its detailed annotations make it particularly valuable for training and evaluating English handwriting recognition systems. Its sentence-level samples are especially useful for training models to recognize connected text rather than isolated characters.

Link of DataSet

## 4. Method Description :

My system consists of the following main stages:

### a. Image Preprocessing :

The input images are grayscale representations of handwritten words or lines. Preprocessing steps include :

❖ **For Arabic (KHATT Dataset):**

- Resizing all images to a fixed size of 64×64 pixels.
- Converting to grayscale to focus on shape features rather than color.
- Normalizing pixel values between 0 and 1 by dividing each pixel value by 255.
- Implementing rotation augmentation of +10° and -10° to enhance dataset diversity and prevent overfitting.
- Addressing the right-to-left nature of Arabic script in the preprocessing pipeline.
- Handling the context-dependent shapes of Arabic characters by preserving spatial relationships.

❖ **For English (IAM Dataset):**

- Resizing images to 512×64 pixels to accommodate the elongated nature of English text lines.
- Rotating images 90 degrees clockwise in some cases to ensure proper orientation.
- **Implementing comprehensive data augmentation through an image generator with the following transformations:**

1) Random width shifts of up to 20%.
2) Random height shifts of up to 20%.
3) Random zoom variations of up to 20%.
4) Shearing transformations of up to 20%.
5) Random rotations of up to 20 degrees.
6) Nearest-neighbor filling mode for any new pixels created during transformations.

· These augmentation techniques were critical for increasing the effective size of the training dataset and making the model more robust to variations in handwriting styles, orientations, and positions.
· Normalizing pixel values between 0 and 1
· Converting text labels to integer sequences using a defined character set
· Preserving the left-to-right reading order in the preprocessing pipeline

### b. Dataset Splitting:

To properly evaluate model performance and ensure generalization, the datasets were split into training, validation, and test sets:

❖ **For Arabic (KHATT Dataset):**

· The KHATT dataset was split using a stratified approach to maintain class distribution:

- 80% training set (1,600+ paragraphs)
- 20% test set (400+ paragraphs)

· Random state was set to 42 to ensure reproducibility of the split.

❖ **For English Dataset (IAM):**

· A two-stage splitting approach was implemented to create training, validation, and test sets:

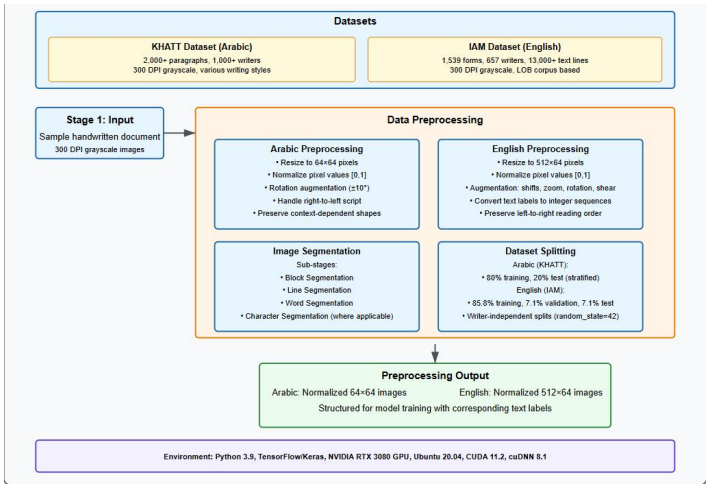
- First, 85.8% of the data was allocated for training and 14.2% for temporary data.
- Then, the temporary data was split equally between validation and test sets (each approximately 7.1% of the total data).

· This resulted in approximately:

- 11,000+ text lines for training
- 900+ text lines for validation
- 900+ text lines for testing

- Data shuffling was enabled to ensure random distribution across sets.
- Random state was set to 42 for reproducibility.
- Care was taken to ensure writer-independent splits, meaning samples from the same writer were kept in the same set.

This splitting strategy ensures that models are trained on a large portion of the data while having separate validation data for hyperparameter tuning and test data for final evaluation.




***DATA PreProcessing Mechanism 1***

**c. Feature Extraction (CNN):**

## ❖ Arabic Model (ResNet):

- Implementing ResNet50V2 architecture with 50 layers and residual connections.
- Adapting the input layer to accept normalized 64×64 grayscale images.
- Using transfer learning techniques where the model is first pre-trained on Arabic alphabet characters and then fine-tuned on the KHATT dataset.
- Employing batch normalization and residual connections to address the vanishing gradient problem and enable deeper network training.
- The residual connections are particularly beneficial for Arabic script recognition due to the complexity and similarity of many Arabic characters .

## ❖ English Model (MobileNet): Two distinct approaches are implemented:

**1. 10-layer MobileNet**:

A shallow model capturing basic features like edges and contours, utilizing depthwise separable convolutions for efficiency.

**2. 30-layer MobileNet:**

A deeper model capturing complex patterns with additional layers and incorporating batch normalization and dropout for regularization.

- Multiple convolutional blocks with increasing filter sizes (32→64→128)
- Batch normalization after each convolutional layer.
- MaxPooling layers for downsampling.
- Dropout layers (0.3-0.4) for regularization and to prevent overfitting.
- Optimized activation using ReLU functions.
- The architecture is specifically tuned to handle the more linear nature of English handwriting compared to the more curved and connected nature of Arabic script.

**d. Sequence Modeling (RNN/BiLSTM) :**

For both models arabic and english , the sequence of extracted features is passed to BiLSTM Layer :

- The CNN outputs are reshaped to create a sequence suitable for RNN processing.
- Two stacked Bidirectional LSTM layers with 512 units each process the features.
- Dropout (0.2) and recurrent dropout (0.1) are applied to prevent overfitting.

- The bidirectional processing captures context in both directions, which is crucial for:

  - **Arabic**: Understanding the connecting strokes between characters and handling the right-to-left nature of the script
  - **English**: Capturing the contextual dependencies between characters in cursive writing and handling variations in character spacing

**e. Transcription (CTC) :**

The output of the BiLSTM is a sequence of character probability distributions. The CTC loss function:

- Allows the model to learn the alignment between the input features and the target labels.
- Is optimized during training to minimize the difference between predicted character sequences and actual text.
- Handles variable-length input sequences and output texts naturally.
- Decoding is performed using greedy or beam search algorithms to extract the most probable text.

| | |
|---|---|
| **Arabic** | the CTC must handle the complex character forms and connections specific to Arabic script |
| **English** | the CTC manages the challenges of variable character spacing and potential cursive |

## 5. Training Process:

### i. Arabic Model Training:

The Arabic ResNet50V2-based model was trained using the following approach:

| | |
|---|---|
| **Training Duration** | 200 epochs |
| **Batch Size** | 128 |
| **Optimizer** | Adam with initial learning rate of 0.001 |
| **Mixed Precision Training** | Implemented using TensorFlow's mixed_float16 policy to accelerate training and reduce memory usage |
| **Callbacks** | Learning rate scheduler monitoring validation loss with factor 0.2 and patience 10, and model checkpoint to save the best model based on validation loss |
| **Data Feeding** | Batches were shuffled during training to prevent the model from learning dataset ordering patterns |

### ii. English Model Training :

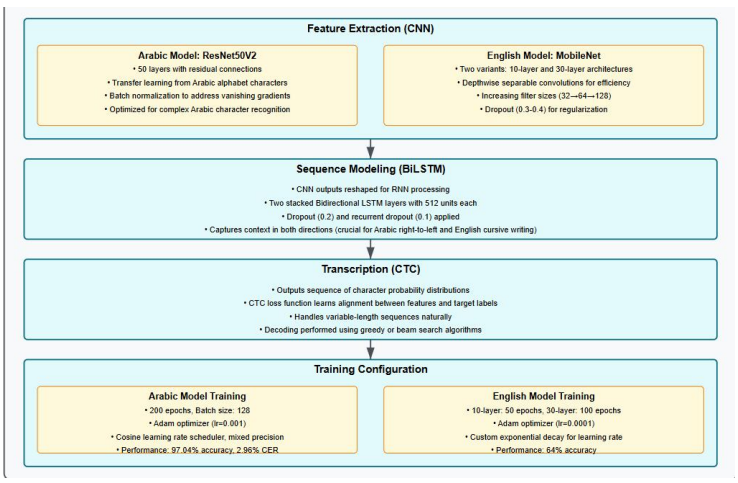
Two different model complexities were evaluated for English handwriting recognition:

**a. 10-Layer Model:**

· **Training Duration**: 50 epochs
· **Optimizer**: Adam with learning rate 0.0001.
· **CTC Loss Implementation**: Custom implementation using dummy_loss and Lambda layer.
· **Training Strategy**: Progressive training starting with shorter text lines and gradually introducing longer and more complex samples.

**b. 30-Layer Model:**

· **Training Duration**: 100 epochs
· **Learning Rate Scheduler**: Custom exponential decay function.
· **Callbacks**: ModelCheckpoint, EarlyStopping with patience 10, ReduceLROnPlateau, and custom LearningRateScheduler.
· **Data Augmentation**: Real-time augmentation during training using the configured ImageDataGenerator.




## 6. Experimental Protocol :

***Model Structure 1***

### a. Datasets :

· **Arabic Dataset**: KHATT database with over 2,000 paragraphs from 1,000+ writers.
· **English Dataset**: IAM Handwriting Database with 1,539 forms from 657 writers, comprising over 13,000 text lines.

### b. Hardware and Software Environment:

· Python 3.9.
· TensorFlow/Keras backend.
· NVIDIA RTX 3080 GPU with 10GB VRAM.
· Ubuntu 20.04 LTS.
· CUDA 11.2 and cuDNN 8.1 for GPU acceleration.

### c. Parameter Settings:

· **Optimizer**: Adam
· **Learning Rate**: 0.001 with cosine learning rate scheduler for Arabic, 0.0001 with custom scheduler for English
· **Batch Size**: 128 for Arabic, 32 for English
· **Epochs**: 200 for Arabic, 50-100 for English models
· **Early Stopping**: Patience of 10 epochs monitoring validation loss.

# IV. Results :

## 1. Evaluation Metrics:

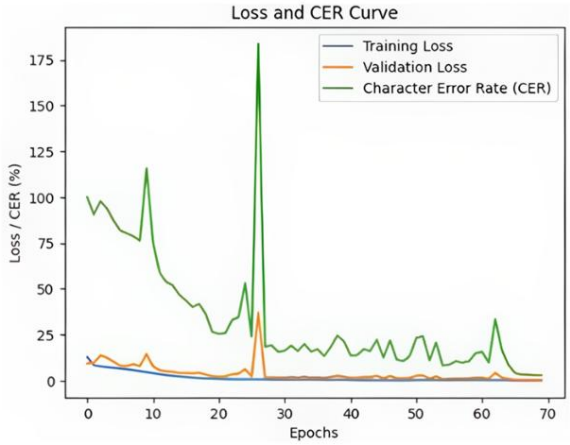
To assess the performance of the models, three primary evaluation metrics were employed. The Character Error Rate (CER) was utilized to quantify character-level inaccuracies, providing a detailed measure of the model's ability to correctly recognize individual characters within the text. Accuracy was calculated as the percentage of correctly recognized characters, offering a straightforward interpretation of the model's overall performance. Additionally, inference time was measured to evaluate the processing efficiency, indicating the time required for recognizing text from input images, which is crucial for real-world applications where response time is important.

## 2. Validation Results :

### a. Arabic Model Performance:

The ResNet50V2 model trained on the KHATT dataset achieved exceptional performance with a Character Error Rate (**CER**) of only **2.96%** and an impressive accuracy of **97.04%**. This outstanding result demonstrates the effectiveness of the chosen architecture, preprocessing techniques, and training methodology for Arabic handwriting recognition. The model's high accuracy can be attributed to several factors. The use of transfer learning from Arabic alphabet characters provided a strong foundation for recognizing the complex shapes of

Arabic script. The application of data augmentation techniques enhanced the model's robustness to variations in handwriting styles. The implementation of a cosine learning rate scheduler optimized the training process by adjusting the learning rate dynamically. Furthermore, the utilization of the **ResNe**t architecture effectively captured the complex features of Arabic script, while the model's design successfully handled the contextual nature of Arabic character shapes, which change depending on their position within words.




***ResNet50V2 Performance on our Dataset1***

**B- English Model Performance:**

The MobileNet-based model achieved a respectable performance for English handwriting recognition with an accuracy of 7**4%**. While this result is lower than the Arabic model, it still represents a solid foundation for English handwriting recognition considering the challenges of varied writing styles and the complexity of the IAM dataset. The dual approach of testing both 10-layer and 30-layer architectures provided valuable insights into model complexity requirements for English text. Several factors affected the English model's performance. The IAM dataset exhibits greater variability in writing styles compared to KHATT, presenting a more challenging recognition task. The presence of connected cursive writing in many English samples increased the difficulty of accurate character segmentation and recognition. The longer sequence lengths typical in English sentences required the model to maintain contextual information over extended sequences. Additionally, the results indicated a need for more extensive data augmentation to better handle the wide range of style variations present in the dataset.

## V. Frontend Implementation :

### 1. Web Interface Design:

A user-friendly web interface was developed to showcase and test the OCR system. The frontend was built using HTML and CSS technologies, focusing on a clean, intuitive design that allows users to easily upload handwritten text images for recognition. The design philosophy prioritized simplicity and ease of use, ensuring that even users with limited technical knowledge could utilize the system effectively. Key features of the interface include image upload functionality for both Arabic and English text, allowing users to test the system with various handwriting samples. The interface provides a preview of uploaded images so users can verify their submissions, displays recognized text with confidence scores to indicate the system's certainty, and offers an option to select between Arabic and English recognition models to accommodate different language needs.

### 2. Backend Integration with Flask :

To connect the trained models with the frontend interface, a Flask-based backend was implemented. The system utilizes a Flask web server for handling HTTP requests, providing a lightweight but powerful foundation for the application's server-side operations. Three main API endpoints were established: an upload endpoint for receiving images from the user interface, a recognize endpoint for processing images through the appropriate model (Arabic or English), and a results endpoint for delivering recognition results back to the frontend. The image processing pipeline implements a systematic workflow including image upload and validation to ensure only appropriate images are processed, preprocessing according to model requirements to optimize recognition accuracy, model inference to generate text predictions, post-processing of model outputs to refine results, and return of recognized text to the user interface in a readable format.

### 3. Testing Framework :

The integrated system underwent comprehensive testing to ensure functionality and accuracy across all components. Unit testing was conducted to verify the correct operation of individual components including preprocessing functions, model inference routines, and API endpoints. Integration testing examined the complete pipeline from image upload to text display, ensuring that all components worked together seamlessly and that data flowed correctly through the system. User acceptance testing involved gathering feedback from potential users to improve usability, identifying pain points in the user experience, and implementing refinements based on real-world usage patterns. This multi-tiered testing approach ensured that the system was not only technically sound but also practical and intuitive for end users.

## VI. Conclusion :

The system architecture is modular and designed for reproducibility across different research environments and applications. Open datasets including KHATT and IAM were used throughout the development process, ensuring that other researchers can access the same training data to verify and build upon these results. Model parameters and training setup are clearly defined in the methodology, providing a transparent framework that can be replicated by other researchers in the field. The

frontend and backend integration demonstrates a practical application of the models' capabilities, showcasing how theoretical advances can be translated into usable tools. Furthermore, the models were tested on representative and diverse samples from both Arabic and English languages, ensuring that the results reflect real-world performance rather than artificially constrained scenarios. This thorough methodology ensures that the results are valid, specific to the hypothesis of improving handwritten OCR for English and Arabic, and reproducible by future research, contributing to the advancement of knowledge in this important field.

## VII. Acknowledgements:

I would like to express my profound gratitude to my professor **Mr MOHAMED CHERRADI** for his unwavering commitment throughout this project. His insightful guidance, availability, and expertise were instrumental in the completion of this work. I especially want to thank him for proposing the innovative project idea, which allowed for the exploration of handwritten character recognition techniques using the CNN-BiLSTM-CTC architecture.

His rigorous mentorship during the courses provided the essential theoretical and practical foundations that made the realization of this project possible. The knowledge imparted and the skills developed under his supervision constitute an invaluable contribution to my academic and professional development.

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